Maharashtra State Board Of Technical Education, Mumbai

Learning and Assessment Scheme Of Diploma Courses for Working Professionals

Programme Name	**: Diploma In Electronics & Tele-communication Engg. (Working Professional)**		
Programme Code	**: PEJ**	**With Effect From Academic Year**	**: 2024-25**
Duration of Programme	**: 6 Semester**		
Semester	**: Fourth**	**Scheme**	**: F**

Sr No	Course Title	Abbrevation	Course Type	Course Code	Diploma Awarding Course	Total IKS Hrs for Sem.	Learning Scheme: Actual Contact Hrs./Week			Learning Scheme: Self Learning (Activity/ Assignment /Micro Project)	Learning Scheme: Notional Learning Hrs /Week	Credits	Assessment Scheme: Paper Duration (hrs.)	Theory				Based on LL & TL: Practical				Based on Self Learning: SLA		Total Marks
							CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
														Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
(All Compulsory)																								
1	PRINCIPLES OF ELECTRONIC COMMUNICATION	PEC	DSC	313326	No	1	3	-	2	1	6	3	3	30	70	100	40	25	10	25@	10	25	10	175
2	MICROCONTROLLER & APPLICATIONS	MAA	DSE	314328	No	0	3	-	4	1	8	4	3	30	70	100	40	25	10	25#	10	25	10	175
3	BASIC POWER ELECTRONICS	BPE	DSC	314363	No	0	3	-	2	1	6	3	3	30	70	100	40	25	10	25@	10	25	10	175
4	ELECTRONIC MEASUREMENTS & INSTRUMENTATION	EMI	AEC	313012	No	0	2	-	2	2	6	3	-	-	-	-	-	50	20	25@	10	25	10	100
Total						**1**	**11**	**0**	**10**	**5**		**13**		**90**	**210**	**300**		**125**		**100**		**100**		**625**

Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA -Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :
1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project./Community (INP) , AbilityEnhancement Course (AEC) , Skill Enhancement Course (SEC) , GenericElective (GE)

Programme Name/s	**: Digital Electronics/ Electronics & Tele-communication Engg./ Electronics & Communication Engg./ Electronics Engineering/ Industrial Electronics**
Programme Code	**: DE/ EJ/ ET/ EX/ IE**
Semester	**: Third**
Course Title	**: PRINCIPLES OF ELECTRONIC COMMUNICATION**
Course Code	**: 313326**

I. RATIONALE

In the fastest growing telecommunication era, diploma engineers deal with electronic communication systems. The use of basic principles of electronic communication will help the students in handling communication systems in the industry and consumer market .This course is developed to empower the students to apply their knowledge and skill to maintain the communication systems.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to attend the following industry/employer expected outcome through various teaching learning experiences .
Maintain basic electronic communication system.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Select relevant frequency range/band for different communication mode.
- CO2 - Maintain AM based communication system.
- CO3 - Maintain FM based communication system.

- CO4 - Identify propagation mode for specified radio frequency band.
- CO5 - Identify relevant type of antenna for given frequency range/application.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme												
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
313326	PRINCIPLES OF ELECTRONIC COMMUNICATION	PEC	DSC	3	-	2	1	6	3	3	30	70	100	40	25	10	25@	10	25	10	175

Total IKS Hrs for Sem. : 1 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
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5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
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V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.

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1	TLO 1.1 Identify the relevant frequency band of electromagnetic spectrum for the specified signal. TLO 1.2 Differentiate analog and digital signal. TLO 1.3 Compare features of the given types of transmission modes. TLO 1.4 Distinguish between external and internal noise sources.	**Unit - I Basics of Electronic Communication** 1.1 Electromagnetic (EM) wave spectrum, frequency bands and their applications 1.2 Signals and its representation: analog and digital signal (In time and frequency domain) 1.3 Block diagram of Analog communication system and operation of each block 1.4 Transmission modes: Simplex , half duplex and full duplex, quadraplex ,Synchronous, Asynchronous and iso-synchoronus 1.5 Noise, Sources of Noise, signal to noise ratio(S/N) and noise figure unit for noise (only concept) 1.6 Ancient communication method in India :-In the history of communication humans relied on non-verbal forms of communication such as drum sounds, pigeons, messengers , symbols and smoke signals. (IKS-1 hour, No question in theory paper)	Chalk-Board Presentations Video Demonstrations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Define modulation and give its need. TLO 2.2 Describe with sketches the given parameters of AM signal. TLO 2.3 Calculate modulation index and modulated signal power of the given AM signal. TLO 2.4 Describe working of the given type of AM detection method . TLO 2.5 Describe working of each block of super heterodyne receiver.	**Unit - II Amplitude Modulation (AM) Communication** 2.1 Modulation ,Need for modulation 2.2 Types of modulation techniques ,Amplitude Modulation: Modulating signal , Carrier signal , modulation Index , mathematical representation of AM Signal (no derivation, only numericals) ,representation in time and frequency domain , Frequency Spectrum ,Types of AM band spectrum (DSB, SSB and VSB)and their applications , Power relations in AM wave (no derivation, only numerical) 2.3 Generation of AM: Block Diagram and working Low level and High Level AM transmitter 2.4 Demodulation of AM signal: Diode detector and practical diode detector, Automatic gain control and Delayed AGC 2.5 Block diagram and working of each block of super heterodyne receiver with waveforms Characteristics Selectivity , Sensitivity and Fidelity	Chalk and Board Videos demostration Visit to Telecommunication Industry

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Differentiate between AM and FM. TLO 3.2 Generate FM signal using given type of method. TLO 3.3 Explain working of the each block of FM receiver . TLO 3.4 Explainworking of the given FM detector circuit. TLO 3.5 Differentiate between FM and PM waveforms.	**Unit - III Frequency Modulation (FM) Communication** 3.1 Frequency modulation: Mathematical representation of FM signal Use of Bessel's function (no derivation),representation of FM signal in time domain and frequency domain, frequency deviation ratio, modulation index (numerical), types of frequency modulation (Narrow Band and Wide Band FM), Concept of Pre-emphasis and De-emphasis and working 3.2 Generation of FM using direct method (varactor diode and reactance modulator) and indirect method (Armstrong method) 3.3 FM detector circuits: Ratio detector and PLL as FM demodulator 3.4 FM Receiver : Block diagram and working with waveforms 3.5 Phase Modulation : definition, waveforms and applications	Chalk and Board Video Demonstrations, Visit to Telecommunication Industry

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Describe with sketches propagation mode of the given type of radio wave /band. TLO 4.2 Describe properties of the given type of wave propagation. TLO 4.3 Define critical frequency, skip distance, skip zone, fading, multiple hop with respect to sky wave propagation. TLO 4.4 Describe the concept of Duct propagation and troposphere scatter propagation of waves.	**Unit - IV Wave Propagation** 4.1 Concept of propagation of radio waves 4.2 Ground Wave propagation 4.3 Sky wave: Ionospheric layers, concept of actual height and virtual height, critical frequency, skip distance, skip zone. concept of fading, maximum usable frequency, multiple hop sky wave propagation 4.4 Space Wave propagation : line of sight, multipath space wave propagation , radio horizon, shadow zone 4.5 Duct propagation (microwave space-wave propagation) Troposphere scatter propagation	Chalk-Board Collaborative learning

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Describe with sketches the working principle of the given type of antenna. TLO 5.2 Sketch the radiation pattern of resonant and non-resonant antenna. TLO 5.3 Compare given type of antenna on the basis of parameters. TLO 5.4 Select type of antenna for transmission and reception of given frequency band.	**Unit - V Antenna** 5.1 Antenna fundamentals :Resonant antenna and Non-resonant antennas, ideal antenna, principle of transmitting and receiving antenna Concept of Electromagnetism interference (EMI) and Electromagnetic Compatibility (EMC) 5.2 Antenna parameters : Radiation pattern , polarization, bandwidth, beam width, antenna resistance, directivity and power gain, antenna gain 5.3 Dipole antenna: Half dipole antenna (Resonant Antenna) and its radiation pattern, folded dipole antenna and its radiation pattern, radiation pattern of Dipole antenna for different length Major lobe , minor lobe , Radiation Pattern for Unidirectional , bidirectional and Omni directional antenna 5.4 Antenna (working principle, constuction, radiation pattern and applications) : Loop antenna, Telescopic antenna, Yagi-Uda antenna, Micro wave antenna - Dish antenna, Horn antenna and Micro-strip patch antenna , Printed antenna, Smart antenna flexible antenna	Chalk-Board Site/Industry Visit Demonstration

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs

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LLO 1.1 Test the output of simplex and duplex mode of communication . LLO 1.2 Determine the number of channels for simplex, half duplex and full duplex Communication .	1	* Demonstrate the simplex ,half duplex and full duplex communication link using switches, wires and LEDs.	2	CO1
LLO 2.1 Test analog and digital signals on CRO and spectrum analyzer. LLO 2.2 Observe the difference between time domain and frequency domain representation of a signal .	2	Observe the different analog and digital signals on CRO and spectrum analyzer in time domain and frequency domain.	2	CO1
LLO 3.1 Interpret the effect of change in modulating frequency on AM signal.	3	Observe the AM modulated waveforms generated for different modulating frequencies.	2	CO2
LLO 4.1 Calculate modulation index 'm' from the observed AM waveform . LLO 4.2 Interprete the effect of 'm' on AM signal.	4	*Generate AM wave and measure its modulation index for different values of modulating signal amplitude.	2	CO2
LLO 5.1 Observe the AM demodulated signal on DSO/CRO. LLO 5.2 Observe the output waveform of AM demodulator and measure its frequency.	5	Build and test the AM demodulator circuit .	2	CO2
LLO 6.1 Observe the AM signal using simulation software. LLO 6.2 Interpret the demodulated AM signal using simulator software.	6	* Display the AM modulator and demodulator signal using MATLAB Simulink/SCILAB /relevant software for different modulating frequencies.	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 7.1 Observe the waveforms and measure the voltages at various test points of AM receiver. LLO 7.2 Troubleshoot various faults of AM receiver such as low volume, hum sound.	7	Test the output of various stages/blocks of the AM receiver.	2	CO2
LLO 8.1 Calculate modulation index 'm' from the observed FM wave . LLO 8.2 Interpret the frequency deviation and modulation index of the FM signal.	8	* Build and test FM signal using voltage controlled oscillator / IC 566 to measure frequency deviation and modulation index.	2	CO3
LLO 9.1 Interpret the FM signal using simulation software. LLO 9.2 Calculate the modulation index and frequency deviation of FM signal.	9	Display FM signal using suitable simulation software such as MATLAB/SCILAB/ relevant software.	2	CO3
LLO 10.1 Build a FM detector circuit using IC 564/565. LLO 10.2 Observe the FM demodulated signal and draw the input and output waveforms.	10	* Demodulate the given FM signal using IC 564/565 and test the output from the given input waveforms.	2	CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 11.1 Observe the waveforms and measure the voltages at various test points of FM receiver. LLO 11.2 Troubleshoot various faults in AM receiver such as popping, hissing etc.	11	Test the output of various stages/blocks of the FM receiver.	2	CO3
LLO 12.1 Interpret the MUF for the given critical frequency.	12	Use simulation software to measure MUF for the given critical frequency and incident angle.	2	CO4
LLO 13.1 Use RF source and field meter to measure the field strength of given antenna. LLO 13.2 Plot the radiation pattern of given antenna .	13	*Test the performance of given dipole antenna ,measure the field strength and plot the radiation pattern for different length of antenna.	2	CO5
LLO 14.1 Use RF source and field meter to measure the field strength of given yagi-uda antenna. LLO 14.2 Plot the radiation pattern of Yagi- Uda antenna and interpret the beamwidth .	14	Test the performance of given Yagi-Uda antenna .	2	CO5
LLO 15.1 Interpret the directivity and beamwidth from the radiation pattern of the given antenna.	15	* Use suitable simulation software to plot the radiation pattern of given antenna.	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Visit

- BSNL/Radio station and prepare report on technique used for modulation demodulation.

Micro project

- 1.Develop a intercom circuit.
2.Build a Walkie Talkie Circuit.
3.Build a circuit of AM receiver.
4.Build a circuit of FM receiver.

Assignment

- 1.Prepare report on AM FM transmission of nearby your city.
2.Draw neat sketches of AM and FM receivers.
3.Draw neat sketches of different AM and FM generation circuits.
4.Prepare Internet based report on the role of electronic communication used for Defense/ISRO.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Digital multimeter : 3 1/2 digit display ,9999 counts digital multimeter measures :Vac,Vdc (1000 V max) ,Adc,Aac (10 A max) , Resistance (0 to 100 Mohm).	1,7,11
2	Antenna trainer kit : for dipole and Yagi- Uda antenna, mobile antenna, omnidirectional antenna, horn antenna, other common types of antenna.	13,14
3	Function generator : Frequency range 0.1 Hz to 30 Mhz.	2,3,4,5,8,10
4	AM Trainer kit for Modulation and Demodulation.	3,4,5,7
5	RF signal generator with wide frequency range 100 Khz to 150 Mhz fine frequency adjustment by calibrated dial built in audio frequency generator.	3,4,5,7,8,10,11

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
6	DSO with Bandwidth : 50-100 MHz TFT colour LCD Dual channel real time sampling 1GSa/s equivalent sampling 25 GSa/s Memory 1Mbpts 10 waveforms and 10 Set ups can be stored.	3,4,5,8,10
7	Simulation software suitable for communication experiments : MATLAB,SCILAB,TINA PRO or any other relevant open source software.	6,9,12,15
8	FM Trainer kit for Modulation (using voltage controlled Oscillator / IC 566) and Demodulation (using IC 564/565) .	8,10,11
9	Cathode ray oscilloscope Dual Trace 20/30/100 Mhz,1 Mega ohm input impedance .	All
10	Regulated power supply : DC supply voltages Dual DC 0-30V ; 0-2A Automatic overload(current protection) constant voltage and constant current operation.	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Basics of Electronic Communication	CO1	6	4	4	4	12
2	II	Amplitude Modulation (AM) Communication	CO2	13	4	6	8	18
3	III	Frequency Modulation (FM) Communication	CO3	10	2	6	6	14
4	IV	Wave Propagation	CO4	7	2	4	6	12
5	V	Antenna	CO5	9	4	4	6	14
Grand Total				**45**	**16**	**24**	**30**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Two offline unit tests of 30 marks and average of two-unit test marks will be consider for out of 30 marks.

For formative assessment of laboratory learning 25 marks.
Each practical will be assessed consider 60% weightage to process, 40% weightage to product.

Summative Assessment (Assessment of Learning)

- End semester assessment of 70 marks.

End semester summative assessment of 25 marks for laboratory learning.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	-	1	1	2	1	1			
CO2	2	3	3	2	2	1	2			
CO3	2	3	3	2	2	1	2			

CO4	2	-	1	2	2	-	1			
CO5	2	-	1	2	2	1	1			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Dennis Roddy John Coolen	Electronic Communication	Pearson Education India, New Delhi , ISBN: 978-817758-558-2
2	GEORGE KENNEDY BERNARD DEVIS	ELECTRONICS COMMUNICATION SYSTEM	TATA Mc- Graw Hill, New Delhi, ISBN: 9780071077828
3	Ravi Kumar Jatoth , T. Kishore Kumar , V .V. Mani	Electronics and Communications Engineering	Apple Academic Press ISBN: ISBN-13 978-1774633892
4	Tomasi W.	ELECTRONIC COMMUNICATION SYSTEM	Pearson Education India, New Delhi, ISBN: 9780130221254
5	Constantine A. Balanis	Antenna Theory: Analysis and Design	Wiley-Student edition India, New Delhi, ISBN: 9788126524228
6	Frenzel George:Dravid Bernard:Prasanna SRM	Electronic Communication Systems	Mc-Graw Hill , New Delhi ,ISBN: 9780071077828
7	R. L. Yadava	Antenna and Wave Propagation	EEE, PHI,ISBN : 9788120342910

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.electronicsforu.com/electronics-projects/fm-radio-receiver-using-ic-ta-7640ap	FM radio receiver

Sr.No	Link / Portal	Description
2	https://www.etti.unibw.de/labalive/index/amplitude-modulation/	Amlitude Modulation
3	https://www.etti.unibw.de/labalive/experiment/amtransmitterrecordaudiodemo/	AM transmitter 1 - record audio transmit signal via file
4	https://www.etti.unibw.de/labalive/experiment/amtransmitrecordedsignal/	AM transmitter 2 - transmit recorded signal via USRP
5	https://www.etti.unibw.de/labalive/experiment/fmtransmitterrecordaudiodemo/	FM transmitter
6	https://www.etti.unibw.de/labalive/experiment/fmdemod/	FM receiver
7	https://www.tutorsglobe.com/homework-help/electrical-engineering/fault-in-fm-radio-receiver-71490.aspx	Fault finding in FM radio receiver.
8	https://www.radioandtvhelp.co.uk/help-guides/radio/troubleshooting-interference-to-am-radio#:~:text=Problems%20on%20AM%20radio%20can,and%20could%20indicate%20reception%20problems.	Fault finding in AM radio receiver.
9	https://ijarsct.co.in/Paper10949.pdf	Simulation software to measure MUF for the given critical frequency and incident angle.
10	https://www.ahsystems.com/articles/Understanding-antenna-gain-beamwidth-directivity.php	Antenna parameters .
11	https://kcgcollege.ac.in/Virtual-Lab/Electronics-and-Communication-Engineering/Exp-2/	Frequency Modulation and Demodulation
12	https://vlab.amrita.edu/?sub=3&brch=163&sim=260&cnt=2644	Amplitude Modulation (Simulation experiment)

Sr.No	Link / Portal	Description

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s	**: Automation and Robotics/ Digital Electronics/ Electronics & Tele-communication Engg./ Electrical and Electronics Engineering/ Electronics & Communication Engg./ Electronics Engineering/ Instrumentation & Control/ Industrial Electronics/ Instrumentation/ Electronics & Computer Engg.**
Programme Code	**: AO/ DE/ EJ/ EK/ ET/ EX/ IC/ IE/ IS/ TE**
Semester	**: Fourth**
Course Title	**: MICROCONTROLLER & APPLICATIONS**
Course Code	**: 314328**

I. RATIONALE

Microcontrollers plays a very important role in the design, development of embedded systems. Automation is used in every field of engineering and microcontroller is inbuilt component of these systems. Diploma engineers have to deal with various microcontroller based systems and maintain them. This course will enable the students to develop the skills to use and maintain microcontroller based applications.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to help students to attain the following industry/employer expected outcome through various teaching learning experiences:

• Maintain microcontroller based systems.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Interpret architecture of 8-bit microcontrollers.

- CO2 - Develop program in 8051 in assembly language for the given operation.
- CO3 - Develop program using timers and interrupts.
- CO4 - Interface the memory and I/O peripherals to 8051 microcontroller.
- CO5 - Maintain microcontroller based applications.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

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314328	MICROCONTROLLER & APPLICATIONS	MAA	DSE	3	-	4	1	8	4	3	30	70	100	40	25	10	25#	10	25	10	175

Total IKS Hrs for Sem. : 0 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
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1	TLO 1.1 List the features of 8051 Microcontroller. TLO 1.2 Explain the significance of selection factors while selecting Microcontroller for application. TLO 1.3 Describe the 8051 block diagram. TLO 1.4 Differentiate Microcontroller and Microprocessor for the given parameters. TLO 1.5 Compare Harvard architecture and Von-Neumann architecture. TLO 1.6 Explain functions of each block of 8051 Microcontroller. TLO 1.7 Compare the given derivatives of 8051 Microcontroller.	**Unit - I Microcontroller Overview and 8051 Architecture** 1.1 Features and selection factors for Microcontroller 1.2 Block diagram of 8051 Microcontroller: CPU, input device, output device, memory and buses 1.3 Comparison of Microcontroller and Microprocessor on basis of: Memory, Complexity, Type of Architecture, Cost, Applications, Typical examples of Microcontrollers and Microprocessors 1.4 Architectures of Microcontroller: Harvard , Von Neumann. Concept of pipelining 1.5 8051 Microcontroller: Architecture, Pin Configuration, Memory Organisation, Power saving options 1.6 Derivatives of 8051 (8951, 8031, 8751). Comparison between derivatives	Learning using Chalk-Board Blended Classroom Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Explain the function of the given software development tools. TLO 2.2 Classify addressing modes of 8051 with examples. TLO 2.3 Describe the function of the given instruction with suitable example. TLO 2.4 Explain the use of the given assembler directives with examples. TLO 2.5 Develop simple programs to perform the following operations: Data manipulation, Masking, Stack operation, Branching execution.	**Unit - II 8051 Programming** 2.1 Software Development Cycle: Editor, Assembler, Compiler, Cross-Compiler, Linker, Locator 2.2 Addressing Modes : Immediate, Register, Direct, Indirect, Indexed 2.3 Instruction set :Data Transfer, Arithmetic, Logical, Branching, Machine control and Boolean 2.4 Assembler Directives: ORG, DB, EQU, END, CODE, DATA 2.5 Assembly Language Programming (ALP): Data manipulation, Masking , Stack operation, Branch related programming	Lecture using Chalk-Board Presentations Blended Learning

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Describe the functions of Timer/ Counters, their applications, and modes of Timers. TLO 3.2 Generate the waveforms by using the given mode of Timer. TLO 3.3 Explain the interrupt mechanism with the help of suitable example. TLO 3.4 Explain the operation of given mode for Serial communication. TLO 3.5 Explain I/O Port Programming.	**Unit - III 8051 Timers, Interrupts, Serial and Parallel Communication** 3.1 Configuration and Programming of Timer/Counter using Special Function Registers [SFRs]: TMOD, TCON, THx, TLx, Simple programs to generate the time delays 3.2 Configuration and Programming of interrupts using SFRs: IE, IP 3.3 Serial Communication SFRs: SCON, SBUF, PCON, Modes of serial communication, Simple Programs on serial communication. Serial Communication using MAX 232 3.4 Configuration and Programming of I/O Port : P0, P1, P2, P3	Lecture using Chalk-Board Hands-on Blended Learning
4	TLO 4.1 Interface Input/Output Devices with 8051 microcontroller. TLO 4.2 Interface ADC with 8051 microcontroller. TLO 4.3 Interface DAC with 8051 microcontroller. TLO 4.4 Describe with neat sketch the interfacing of the given external memory. TLO 4.5 Describe the procedure to troubleshoot the given I/O device.	**Unit - IV 8051 Interfacing** 4.1 I/O Interfacing: Keyboard, Relays, LED, LCD, Seven Segment display 4.2 Interfacing ADC 0808/09 with 8051. Simple programs for ADC interfacing 4.3 Interfacing DAC 0808/09 with 8051. Simple programs for DAC interfacing 4.4 Memory Interfacing: Program and Data Memory	Lecture using Chalk-Board Hands-on Blended Learning Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Generate the given waveform using 8051 and DAC. TLO 5.2 Interface Analog Input devices with 8051 microcontroller. TLO 5.3 Program 8051 for the given application. TLO 5.4 Interface Stepper motor to 8051. TLO 5.5 Describe the procedure to troubleshoot the given microcontroller based application.	**Unit - V 8051 Applications** 5.1 Square and Triangular waveform generation using DAC 5.2 Temperature sensor (LM35) interfacing using ADC to 8051 5.3 Water Level controller design using 8051 5.4 Stepper Motor Interfacing to 8051 to rotate in clockwise and anticlockwise direction	Lecture using Chalk-Board Hands-on Blended Learning Presentations

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify the functions of various blocks of 8051 microcontroller development board.	1	* Identification of various blocks of 8051 microcontroller development board	2	CO1
LLO 2.1 Develop an Assembly Language Program (ALP) for addition of two numbers using various addressing modes and assembler directives.	2	Assembly Language Program using various addressing modes	2	CO2
LLO 3.1 Develop an ALP to perform arithmetic operations: addition, subtraction, multiplication and division on 8-bit data.	3	* ALP to perform arithmetic operations on 8-bit data	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 4.1 Develop an ALP to perform arithmetic operations: addition, subtraction on 16-bit data.	4	* ALP to perform arithmetic operations on 16- bit data	2	CO2
LLO 5.1 Develop an ALP to perform addition of BCD data stored at external memory and store result in internal memory.	5	* ALP to perform addition of BCD data	2	CO2
LLO 6.1 Develop an ALP for sum of series of numbers stored in RAM locations 40-49H. Find the sum of the values at the end of the program, store the lower byte in 30H and the higher byte in 31H.	6	* ALP for series addition	2	CO2
LLO 7.1 Develop an ALP to transfer data from source to destination locations of internal/ external data memory.	7	* Array data transfer from source locations to destination locations	2	CO2
LLO 8.1 Develop an ALP to exchange block of data from source to destination location of internal/ external data memory.	8	* Block exchange of data from source locations to destination location	2	CO2
LLO 9.1 Develop an ALP for identifying smallest number from the given data bytes stored in internal/ external data memory.	9	* Finding the smallest number from the given data bytes	2	CO2
LLO 10.1 Develop an ALP for identifying largest number from the given data bytes stored in internal/ external data memory.	10	Finding the largest number from the given data bytes	2	CO2
LLO 11.1 Develop an ALP for arranging numbers in ascending order stored in internal/ external data memory.	11	* Arranging the numbers in ascending order	2	CO2
LLO 12.1 Develop an ALP for arranging numbers in descending order stored in internal/ external data memory.	12	Arranging numbers in descending order	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 13.1 Write an ALP to generate delay using timer register.	13	* Generate delay using timer register	2	CO3
LLO 14.1 Develop an ALP to transfer 8 bit data serially on serial port.	14	* Serial 8 bit data transfer on serial port	2	CO3
LLO 15.1 Interface LED with microcontroller and turn it 'ON' with microcontroller interrupt.	15	LED interfacing to 8051	2	CO4
LLO 16.1 Develop an ALP to generate pulse and square wave by using timer delay.	16	Generating Pulse and Square wave using timer delay	2	CO4
LLO 17.1 Interface 4 X 4 LED matrix with 8051 to display various pattern.	17	LED matrix Interfacing to 8051	2	CO4
LLO 18.1 Interface 7-segment display to display the decimal number from 0 to 9.	18	* Seven Segment Display interface for displaying decimal numbers	2	CO4
LLO 19.1 Interface relay with microcontroller and turn it 'ON' and 'OFF'.	19	* Relay interfacing to Microcontroller	2	CO4
LLO 20.1 Interface LCD with 8051 microcontroller to display the characters and decimal numbers.	20	* LCD interfacing to 8051 to display characters and decimal numbers	2	CO4
LLO 21.1 Interface the given keyboard with 8051 and display the key pressed.	21	Keyboard interfacing to 8051	2	CO4
LLO 22.1 Interface ADC with 8051 microcontroller and verify input/output.	22	* ADC interfacing to 8051	2	CO4
LLO 23.1 Interface DAC with 8051 microcontroller to generate square wave.	23	* DAC Interfacing to generate the square waveform	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 24.1 Interface DAC with 8051 microcontroller to generate triangular wave, saw-tooth wave.	24	DAC interfacing to generate the triangular waveforms	2	CO5
LLO 25.1 Interface stepper motor to microcontroller and rotate in clockwise direction at the given angles.	25	* Stepper Motor interfacing to 8051	2	CO5
LLO 26.1 Interface stepper motor to microcontroller and rotate in anti-clockwise direction at the given angles.	26	Stepper Motor interfacing to 8051 for rotating anti-clockwise	2	CO5
LLO 27.1 Design water level controller using any suitable open source simulation software to detect and control the water level in a tank.	27	Water Level Controller using 8051	4	CO5
LLO 28.1 Interface temperature sensor LM35 to 8051 to read temperature, convert it to decimal and send the value to Port 0 with some delay.	28	Temperature Sensor interfacing to detect and measure temperature	4	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Build a class period bell using microcontroller 8051.
- Build a circuit using 8051 microcontroller to blink LED.

- Build a circuit to display number 0 to 9 with a given delay.
- Build digital clock with 8051 microcontroller.
- Develop Fire Detection System using smoke and temperature sensor.

Student Activity

- Prepare power point presentation on applications of microcontroller.
- Undertake a market survey of different microcontrollers.

Assignment

- Prepare a chart of various features using data sheets of 8051 microcontroller and its derivatives.
- Prepare chart of stepper motor to display its features and steps for its operations using data sheets.
- Prepare a chart of various types of ADC and DAC to display its features and pin functions using data sheets.
- Prepare a chart of various types of LCDs to display its features , pin functions and steps of operations using data sheets.
- Prepare a power point presentation on 8051 interfacing/applications.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	DSO with Bandwidth : 50-100 MHz TFT colour LCD Dual channel real time sampling1GSa/s equivalent sampling 25 GSa/s Memory 1Mbpts 10 waveforms and 10 Set up scan be stored.	13,16,23,24
2	4X4 LED matrix suitable to interface with 8051 trainer kit	17
3	7-segment LED Display	18
4	Relay trainer board suitable to interface with 8051 trainer kit	19
5	LCD trainer board	20
6	Keyboard: 4 x 4 trainer board	21
7	ADC(0808) trainer board	22
8	DAC (0808) trainer board	23,24

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
9	Stepper Motor: 50/100 rpm	25,26
10	Water level controller kit	27
11	Temperature Controller trainer board	28
12	Temperature Sensor LM35: 5V operating voltage, Operating temperature range (°C) -55 to 150, analog output	28
13	8051 Microcontroller kit: On-chip 64 KB ISP+IAP flash, 1KB SRAM, 5V operating voltage, 0 to 40 MHz 64 kB of on-chip Flash program memory	All
14	Desktop PC with microcontroller simulation software.	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Microcontroller Overview and 8051 Architecture	CO1	11	2	6	6	14
2	II	8051 Programming	CO2	8	4	4	4	12
3	III	8051 Timers, Interrupts, Serial and Parallel Communication	CO3	10	4	4	6	14
4	IV	8051 Interfacing	CO4	10	4	6	8	18
5	V	8051 Applications	CO5	6	2	4	6	12
Grand Total				**45**	**16**	**24**	**30**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Two offline unit tests of 30 marks and average of two unit test marks will be consider for out of 30 marks.
- For formative assessment of laboratory learning 25 marks.
- Each practical will be assessed considering 60% weightage to process, 40% weightage to product.

Summative Assessment (Assessment of Learning)

- End semester assessment is of 70 marks.
- End semester summative assessment is of 25 marks for laboratory learning.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	1	1	1	1	-	1			
CO2	2	2	2	2	1	-	2			
CO3	2	2	2	1	1	1	2			
CO4	2	2	2	2	1	-	2			
CO5	2	3	2	2	1	2	2			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Mazidi Muhammad Ali, Mazidi Janice Gillispe, Mckinlay Rolin D	The 8051 Microcontroller and Embedded Systems: Using Assembly and C	Pearson Publication, 2017 ISBN: 9788131710265
2	Ayala Kenneth J	The 8051 Microcontroller	Thomson Delmar Learning, 2004 ISBN: 9781401861582
3	Deshmukh Ajay V	Microcontroller: Theory and Application	McGraw Hill,2011 ISBN: 9780070585959
4	Pal Ajit	Microcontrollers: Principle and Application	PHI Learning, 2014 ISBN: 978812034394
5	Chattopadhyay Santanu	Microcontroller and Applications	All India Council for Technical Education, 2023 ISBN: 9788196057602

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	http://vlabs.iitkgp.ac.in/rtes/#	Keyboard-MCU interfacing take a input from keypad and display on LCD
2	https://studytronics.weebly.com/8051microcontroller.html	8051 Microcontroller Architecture, Internal Memory , Instruction Set, Timers and Counters, Interrupts

Sr.No	Link / Portal	Description
3	https://archive.nptel.ac.in/courses/108/105/108105102/	S. Chattopadhyay, SWAYAM/NPTEL course on “Microprocessors and Microcontrollers”
4	https://www.keil.com/download/product/	Introduction to KEIL tool for 8051 programming
5	https://www.dnatechindia.com/Interfacing-LCD-to-8051.html	Interfacing LCD to 8051
6	https://web.mit.edu/6.115/www/document/8051.pdf	MCS@51 Microcontroller family user’s manual
7	https://econtent.msbte.edu.in/econtent/marathi_econtent.php	Microcontroller and Applications Learning Material In Marathi-English

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s	**: Digital Electronics/ Electronics & Tele-communication Engg./ Electrical and Electronics Engineering/ Electronics & Communication Engg./ Electronics Engineering/ Instrumentation & Control/ Industrial Electronics/ Instrumentation/**
Programme Code	**: DE/ EJ/ EK/ ET/ EX/ IC/ IE/ IS**
Semester	**: Fourth**
Course Title	**: BASIC POWER ELECTRONICS**
Course Code	**: 314363**

I. RATIONALE

Power electronics plays a important role in the efficient use of electrical energy and environmental control. The power electronic circuits are used in industrial automation and in manufacturing sector of control circuits. This course is developed to empower the students to apply their knowledge to solve broad power electronics based industrial application problems.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to help students to attain the following industry/employer expected outcome through various teaching learning experiences:
• Maintain electronic control systems comprising of power electronic components.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Identify power semiconductor devices used in Power Electronics circuit.
- CO2 - Maintain SCR Triggrering and Commutating Circuits.

- CO3 - Use phase controlled rectifiers in different applications.
- CO4 - Analyze power converter circuits.
- CO5 - Maintain power electronic circuits used in various domestic and industrial applications.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
314363	BASIC POWER ELECTRONICS	BPE	DSC	3	-	2	1	6	3	3	30	70	100	40	25	10	25@	10	25	10	175

Total IKS Hrs for Sem. : 0 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Classify Thyristor family devices on the basis of applications and features. TLO 1.2 Explain the working of various power electronics devices with sketches. TLO 1.3 Interpret V-I characteristic of the given power electronic device. TLO 1.4 Calculate latching and holding current for given thyristor. TLO 1.5 Select proper triggering device for the given circuit and justify it. TLO 1.6 Identify various power electronic devices along with their specification for given application.	**Unit - I Power Semiconductor Devices** 1.1 Classification of Thyristor family devices 1.2 Construction ,working principle, V-I characteristics and applications of Power diode ,Power MOSFET and IGBT, Reverse recovery characteristics of power diode 1.3 SCR- Construction ,working principle, V-I characteristics and applications, Two transistor analogy, latching and holding current for SCR 1.4 LASCR, TRIAC, GTO,SCS - Construction ,working principle, V-I characteristics and applications 1.5 Triggering devices :UJT, PUT, SUS, SBS, DIAC - Construction, working principle, V-I characteristics and applications	Presentations Lecture Using Chalk-Board

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Describe the turn- ON mechanism of given SCR circuit. TLO 2.2 Explain with sketches the effect of the given firing angles on load voltages. TLO 2.3 Explain with sketches the triggering methods for the given SCR. TLO 2.4 Differentiate various types of commutation methods for SCR with sketches. TLO 2.5 Justify the need of protection circuit for SCR. TLO 2.6 Explain with sketches the working of protection circuits for the given SCR against over voltage and over current.	**Unit - II Triggering and Commutation methods of SCR** 2.1 Concept of turn ON mechanism for given SCR: High voltage, thermal triggering, dv/dt triggering, gate triggering 2.2 Gate trigger circuits: Types of gate signals: DC signal, AC signal and pulse signal 2.3 Thyristor Triggering Circuits: Resistance Triggering Circuit, Resistor-Capacitor (RC) Triggering Circuit, half wave and full wave triggering Circuit, UJT (Unijunction Transistor) Triggering Circuit, Pulse Transformer Triggering Circuit, UJT/ PUT-relaxation oscillator circuit 2.4 Turn OFF (commutation) methods: Natural and Forced Commutation, Types: Class A, Class B, Class C, Class D, Class E, Class F 2.5 SCR protection circuits: Need, Factors causing permanent damage to SCR, causes of over voltage and over current, Over voltage protection circuits using RC snubber circuit and non linear resistor, over current protection circuit using Fuse operation, Electronic crowbar protection circuit	Presentations Lecture Using Chalk-Board

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Explain with sketches the effect of change in firing angle on output voltage of the given rectifier considering concept of phase control. TLO 3.2 Explain operation of Half wave and Full wave controlled rectifiers for given load. TLO 3.3 Explain operation of Semi-converters for given load. TLO 3.4 Calculate load voltage and load current of the given controlled rectifier. TLO 3.5 Describe working principle of multiphase rectifiers with circuit digram.	**Unit - III Phase controlled rectifiers** 3.1 Phase control parameters: Firing angle , and conduction angle 3.2 Single phase half wave controlled rectifier: circuit diagram, working and waveforms with R and RL load, effect of freewheeling diode with RL load, numerical 3.3 Single phase centre tapped full wave controlled rectifier and Bridge rectifier: circuit diagram, working and waveforms with R and RL load, effect of freewheeling diode with RL load, numerical 3.4 Semi- converters: circuit diagram, working and waveforms with R and RL load, effect of freewheeling diode with RL load 3.5 Three phase rectifier: need, circuit diagram, working and waveforms with R load	Lecture Using Chalk-Board Presentations Video Demonstrations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Explain types of converters and classify. TLO 4.2 Explain the working of the given Choppers with sketches . TLO 4.3 Explain with sketches the working of the given type of inverter circuit. TLO 4.4 Describe performance parameters for inverters. TLO 4.5 Explain with sketches the working of the given type of Cycloconverter.	**Unit - IV Power Converters** 4.1 Chopper: Introduction, classification 4.2 Block diagram and working of step down chopper using IGBT, with R and RL load 4.3 Step up chopper using IGBT with R load 4.4 Inverter: Introduction, classification, Block diagram and working of Series inverter, Parallel inverter, Single phase Half bridge and Full bridge inverter 4.5 Performance parameters for the inverter: Harmonic factor of nth Harmonic, Total Harmonic Distortion, Distortion Factor, Lowest order Harmonic 4.6 Cyclo-converter: Introduction, Classification, Single phase Cyclo-converter: working principle of Midpoint configuration with R load	Lecture Using Chalk-Board Presentations Flipped Classroom

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Describe the use of power electronic device in the given industrial circuit. TLO 5.2 Describe the performance of the given Industrial control circuit. TLO 5.3 Explain with sketches the working of the given type of UPS. TLO 5.4 Explain with sketches the working of the given type of SMPS.	**Unit - V Industrial applications of power electronic devices** 5.1 Proximity detector and Time delay circuit using SCR and PUT/UJT 5.2 Battery charger, Emergency light system and Flasher circuit using SCR 5.3 Static AC and DC circuit breaker and Zero Voltage Switch 5.4 Application of Choppers in Electric vehicles 5.5 Block diagram and concept of Online and Offline UPS 5.6 SMPS: concept, Block diagram and applications	Lecture Using Chalk-Board Presentations

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Test the SCR in forward conduction state and measure holding current (IH) and latching current (IL).	1	*Performance of SCR using IC 2N4103 or any other equvalent IC	2	CO1
LLO 2.1 Test the forward and transfer characteristics of given IGBT.	2	*Performance of IGBT using IC BUP 402 or any other equivalent IC	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Test the performance of DIAC and plot its V-I characterstics.	3	*Performance of DIAC using IC DB3/DB4 or any other equivalent IC through its V-I curve	2	CO1
LLO 4.1 Test the R and RC triggering circuits of SCR.	4	Masurement of output voltage by changing firing angle through variation in resistor, capacitor in R and RC triggering circuits of SCR.	2	CO2
LLO 5.1 Measure output voltage by changing firing angle in synchronized UJT triggering circuit.	5	*Synchronized UJT triggering circuit.	2	CO2
LLO 6.1 Observe and verify Input-Output waveforms of Class C-Complimentary type commutation circuit.	6	*Class C-Complimentary type commutation circuit	2	CO2
LLO 7.1 Observe and verify Input-Output waveforms of half wave controlled rectifier with R, RL load and measure load voltage.	7	* Half wave controlled rectifier	2	CO3
LLO 8.1 Observe and verify the Input-output waveforms of full wave controlled rectifier with R, RL load and measure load voltage.	8	Performance of full wave controlled rectifier with R, RL load and measure load voltage.	2	CO3
LLO 9.1 Calculate firing angle and observe input-output voltage waveforms of 3- phase half wave controlled rectifier using Delta-star transformer.	9	Performance of 3- phase half wave controlled rectifier	2	CO3
LLO 10.1 Measure output voltage of step-up chopper for different values of duty cycles.	10	Performance of step-up chopper for different values of duty cycles	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 11.1 Measure output voltage of step-down chopper for R load. LLO 11.2 Measure output voltage of step-down chopper for RL load.	11	*Step-down chopper for R and RL load	2	CO4
LLO 12.1 Measure frequency and output voltage of parallel inverter.	12	Performance of parallel inverter	2	CO4
LLO 13.1 Simulation of single phase midpoint Cyclo-converter with R load.	13	Single phase midpoint Cyclo-converter with R load.	2	CO4
LLO 14.1 Build / test Light dimmer circuit using DIAC-TRIAC.	14	*Light dimmer circuit using DIAC-TRIAC	2	CO5
LLO 15.1 Build / Test Emergency Light circuit using SCR.	15	Emergency Light circuit using SCR	2	CO5
LLO 16.1 Simulation of Temperature controller using SCR.	16	Temperature controller using SCR	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Build Battery charger circuit for charging a battery of 6V, 4AH.
Build fan speed regulator circuit using DIAC and TRIAC.
Build Speed control circuit for12V DC shunt motor using IGBT.
Build a circuit to control Intensity of light using phase control.
Build a circuit for Automatic street light using SCR.

Assignment

- Make Power point presentation on application of Chopper in Electric vehicle.
Make report on use of power electronics based systems in home/industrial applications.
Make a report on role of power electronic devices/system in application of EV Charging Station.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Trainer kit for SCR: Trainer kit suitable to plot anode - cathode characteristics and gate characteristics of silicon controlled Rectifier (SCR). 0 - 10V and 0 - 150V DC power supply of required current rating & 4 no of digital voltmeter & current meter are inbuilt in the kit.	1
2	Function generator: 1 MHz , sine, square, triangular, ramp and pulse generator Freq range 0.01 Hz to 1 Mhz, Output amplitude 20V open circuited, Output impedance 50 ohms. Facility to indicate output frequency & amplitude on dispay.	10
3	MATLAB-SIMULINK / Scilab software, Proteus software, Multisim software	13,16
4	Trainer kit for IGBT:Trainer kit suitable to plot characteristics of IGBT. 2 no of variable DC Power supply of required current rating & 4 no of digital voltmeter & current meter are built in the kit.	2
5	Trainer kit for DIAC :Trainer kit suitable to plot forward and reverse characteristics of Diac. 0-50V DC power supply & required digital voltmeter & current meter is inbuilt in the kit.	3
6	Trainer kit for SCR triggering circuits :Trainer kit suitable to study the basic triggering methods of SCR like Resistance triggering circuit; R-C triggering circuit; UJT triggering circuit; IC 555 triggering circuit etc. Should be provided with SCR, Lamp load (15W) & isolation transformer. Required R & C components are provided in trainer which can be interconnected by patch cords to make the desired configuration. SCR should be operated on 230V, 50Hz AC supply.	4,5
7	Trainer kit for HWR , FWR without and with Capacitor and Inductor Filter: Trainer kit shall consists of Following parts provided on PCB with connecting terminals & test points. Mains transformer primary 230V A.C. Secondary centre tap 12-0-12VAC at 500 mA. 4 diodes which can be interconnected by patch cords to make HWR, FWR circuit, Filter Choke coil, filter Capacitors, Load Resistors. Required configuration of rectifier and filter can be assembled by patch cords. Waveforms can be observed on CRO & various measurements can be done. Line & load regulation can be found out.	7,8
8	LCR Q meter: Accurate 0.01% up to 5 MHz	8,10,11

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
9	Regulated power supply: 0- 30 Volt, 2 A with digital display, with S.C. protection	All
10	Digital multimeter: 3.5 digit with R , V, I measurements, diode and BJT testing	All
11	CRO : Dual Channel, 4 Trace CRT / TFT based Bandwidth 20 MHz/30 Mhz X10 magnification 20 nS max sweep rate, Alternate triggering ,Component tester and with optional features such as Digital Read out , USB interface	All
12	SCR(IC 2N4103),TRIAC(IC BT 139),MOSFET(IC 47N60C3),IGBT(BUP 402),DIAC (DB3/DB4 SSD3A)any other relevant IC can be used,	All
13	Analog multimeter: Suitable to measure AC/DC voltage , Current and Resistance, to test power devices DC voltage Range 400mV to 1000 V AC Voltage Range 4V to 750 V ,DC current 4 mA to 10A ,AC current 4 mA to 10 A Resistance 400 Ohm to 40 M ohm or any other better specifications and facilities	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Power Semiconductor Devices	CO1	12	4	8	4	16
2	II	Triggering and Commutation methods of SCR	CO2	8	4	4	6	14
3	III	Phase controlled rectifiers	CO3	10	4	4	8	16
4	IV	Power Converters	CO4	9	4	4	6	14
5	V	Industrial applications of power electronic devices	CO5	6	2	4	4	10
Grand Total				**45**	**18**	**24**	**28**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Two offline unit tests of 30 marks and average of two unit test marks will be consider for out of 30 marks.

For formative assessment of laboratory learning 25 marks
Each practical will be assessed considering 60% weightage to process, 40% weightage to product.

Summative Assessment (Assessment of Learning)

- End semester assessment is of 70 marks.

End semester summative assessment of 25 marks for laboratory learning.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	-	1	1	2	1	1			

CO2	3	-	1	1	2	1	1			
CO3	3	2	1	1	1	1	1			
CO4	2	2	2	1	2	1	1			
CO5	2	2	2	2	3	3	1			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	M.D. Singh, K. B .Khanchandani	Power Electronics	Tata Mc,Graw Hill ,ISBN-13: 9780070583894
2	P.S.Bimbhra	Power Electronics	Khanna Publisher, New Delhi, ISBN-10. 9788174092793
3	Rashid, Muhammad H.	Power Electronics Circuits Devices and Applications	Pearson Education India, New Delhi,ISBN-10. 9332584583
4	B.R.Gupta And V.Singhal	Power Electronics	S.K.Kataria and Sons, ISBN 10: 9350141078
5	Harish C Rai	Power electronics and Industrial application	CBS publishers ISBN-13: 9789386827869
6	Robert W.Erickson Dragan Maksimovic	Fundamental of Power Electronics	Springer,ISBN-13: 9783030438791

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description

Sr.No	Link / Portal	Description
1	https://learnabout-electronics.org/Semiconductors/thyristors_63.php	Thyristor family, Thyristor protection
2	https://www.electronics-tutorials.ws/power/unijunction-transistor.html	Thyristor family devices, SMPS, Rectifiers.
3	https://www.electrical4u.com/chopper-dc-to-dc-converter/	Chopper operation
4	https://www.elprocus.com/cycloconverters-types-applications/	Cyclo-Converter
5	https://www.alldatasheet.com/	All Datasheets

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s	**: Digital Electronics/ Electronics & Tele-communication Engg./ Electronics & Communication Engg./ Electronics Engineering/ Industrial Electronics/ Medical Electronics**
Programme Code	**: DE/ EJ/ ET/ EX/ IE/ MU**
Semester	**: Third**
Course Title	**: ELECTRONIC MEASUREMENTS & INSTRUMENTATION**
Course Code	**: 313012**

I. RATIONALE

This course is designed to enable the students to handle different test and measuring instrument for testing various electronics and automation systems. Handling test and measuring instrument is the essential activity in any electronic industry. This course will develop skills to select various types of sensors and transducers and to maintain automation system.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to help students to attain the following industry/employer expected outcome through various teaching learning experiences:
Maintain automation systems.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Interpret the characteristics of measuring instrument.
- CO2 - Use various test and measuring instrument.
- CO3 - Interpret working of various types of sensors and transducers.

- CO4 - Measure physical quantities using various types of transducers and sensors.
- CO5 - Maintain signal conditioning and data acquisition systems.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											Total Marks
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
313012	ELECTRONIC MEASUREMENTS & INSTRUMENTATION	EMI	AEC	2	-	2	2	6	3	-	-	-	-	-	50	20	25@	10	25	10	100

Total IKS Hrs for Sem. : 0 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Select instrument on basis of nature of operation. TLO 1.2 List static and dynamic characteristics of the given measuring instrument. TLO 1.3 Identify standard used of given instrument.	**Unit - I Fundamentals of electronic measurements** 1.1 Fundamentals of measuring instruments 1.2 Static and dynamic characteristics of instrument 1.3 Calibration and standards of Instrument: International, primary, secondary, working	Teacher input Demonstration Hands on practice

ELECTRONIC MEASUREMENTS & INSTRUMENTATION **Course Code : 313012**

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Measure output voltage at each test point of CRO. TLO 2.2 Calculate unknown frequency and phase of given input signal by observing Lissajous pattern. TLO 2.3 Describe the procedure to measure frequency, time period, voltage using DSO. TLO 2.4 Measure output of Function generator for sine, square, triangular waveform. TLO 2.5 List front panel controls of Spectrum Analyzer. Measure fundamental and side band frequency, power of given signal.	**Unit - II Testing and measuring Instrument** 2.1 Cathode Ray Oscilloscope (CRO): Introduction, CRO block diagram, function of each block, uses of CRO:Time and frequency measurement, Voltage measurement, Lissagous patterns for phase and unknown frequency measurement 2.2 Digital Storage Oscilloscope (DSO): Introduction, block diagram, operation and uses of digital storage oscilloscope(DSO):Time and frequency measurement, voltage measurement, store and recall mode 2.3 Function generator: Introduction, block diagram, operation and uses of function generator: Sine, square, Triangular waveforms 2.4 Spectrum Analyzer: Introduction, block diagram, operation and uses of spectrum Analyzer: Power performance of signals	Teacher input Hands-on

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Observe output at each block of Instrumentation system. TLO 3.2 Differentiate sensors and transducers. TLO 3.3 Suggest transducer for particular application. TLO 3.4 Describe working of thermal, optical, magnetic and electric sensors. TLO 3.5 Select thermal, optical, magnetic and electric sensors for given application.	**Unit - III Sensors and Transducers** 3.1 Instrumentation System: Block diagram of Instrumentation system, function of each block 3.2 Sensors and Transducer: definition, difference between sensors and transducers ,classification of sensors 3.3 Thermal, optical, magnetic and electric sensors: working principle and applications 3.4 Transducer: Need for Transducer, selection criteria of transducer, types: primary and secondary, active and passive, analog and digital, resistive, capacitive, inductive (Linear variable differential transformer (LVDT), Rotary variable differential transformer (RVDT), Piezo electric transducer	Demonstration Presentations Flipped Classroom

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Measure temperature by selecting transducer. TLO 4.2 Describe procedure to measure pressure using Bourdon Tube, Bellows, Diaphragm. TLO 4.3 Suggest the flow meter to measure flow of the fluid. TLO 4.4 Measure humidity using hygrometer and pH value by pH meter.	**Unit - IV Application of Sensors and Tranducers** 4.1 Temperature measurement types: Resistance Temperature Detector (RTD)– (PT-100) , Thermistors, Thermocouple – Seeback & Peltier effect , Type J, K, R, S, T etc. (Based on material, temperature ranges) , Pyrometer – Optical type 4.2 Pressure measurement types: Bourdon Tube, Bellows, Diaphragm 4.3 Flow measurement types: variable head flow meter, venturimeter, orifice plate. Variable area flow meter: Rotameter, electromagnetic flow meter 4.4 Special transducers and measurement: Humidity measurement using hygrometer, pH measurement	Flipped Classroom Demonstration Hands on practice
5	TLO 5.1 Identify signal conditioning block in any instrumentation system. TLO 5.2 Describe the Data acquisition system.	**Unit - V Data Acquisition System** 5.1 Signal conditioning: Introduction, types, block diagram and working of AC and DC signal conditioning circuits 5.2 Data Acquisition Systems (DAS): Introduction, block diagram, working and applications of DAS	Teacher input Demonstration Site/Industry Visit

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Determine accuracy, resolution, hysteresis of analog multimeter.	1	* Test performance of analog multimeter.	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 2.1 Use CRO to measure amplitude and frequency of the given input signal.	2	Measurement of amplitute and frequency of given signal using CRO.	2	CO2
LLO 3.1 Display Lissajous pattern on CRO and interpret it to measure frequency of the given input signal.	3	* Lissajous pattern to measure unknown frequency.	2	CO2
LLO 4.1 Display Lissajous pattern on CRO and interpret it to measure phase of the given input signal.	4	Lissajous pattern for phase measurement.	2	CO2
LLO 5.1 Measure amplitude and frequency of the given input signal using DSO.	5	* Measurement of amplitude and frequency using DSO.	2	CO2
LLO 6.1 Use spectrum analyzer to measure frequency band of the given input signal.	6	* Frequency band measurement using Spectrum analyzer.	2	CO2
LLO 7.1 Test the performance of the given potentiometer.	7	Test potentiometer characteristics.	2	CO3
LLO 8.1 Measure Linear displacement using LVDT.	8	* Linear displacement measurement using LVDT.	2	CO3
LLO 9.1 Measure pressure using strain gauge.	9	Pressure measurement using strain gauge.	2	CO4
LLO 10.1 Measure temperature of the given liquid using RTD (PT-100) .	10	Temperature measurement of of the given liquid using RTD.	2	CO4
LLO 11.1 Measure temperature using thermocouple (J or K type).	11	* Temperature measurement using thermocouple.	2	CO4
LLO 12.1 Measure flow of fluid using venturi meter .	12	* Flow measurement using venturi tube.	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 13.1 Measurement of flow using orifice plate .	13	Flow measurement using orifice plate.	2	CO4
LLO 14.1 Use rotameter to measure flow of liquid.	14	Flow measurement using Rotameter.	2	CO4
LLO 15.1 Use pH meter to measure pH value of given solution.	15	* pH measurement using pH meter.	2	CO4
LLO 16.1 Interpret the performance of Portable Data Acquisition System.	16	Test the performance of each block of Data Acquisition System.	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Measure temperature of hot plate using semiconductor temperature sensor (LM35).
- Use a fire sensor to make a small electronic alarm circuit.
- Make temperature control circuit using thermistor.
- Make a small circuit using LDR as a sensing device.

- Design D.C. signal conditioning circuit using Wheatstone bridge circuit and implement on PCB.

Assignment

- Prepare a power point presentation on Spectrum analyzer.
- Prepare a chart for use of DSO in various industries.
- Prepare a chart to show the uses of DSO.
- Make a report on use of Spectrum analyzer in Telecommunication field.

Field visit

- Visit to nearby Industry to observe working of various instruments and prepare a detail report.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Analog multi-meter: 0-10A, 0-600V, 0-10 Mohms	1
2	RTD: Pt 100	10
3	Thermocouple:Type: J or K type	11
4	Venturi tube: 12" X 4.35" or equivalent, wide range of diameter ratios, tube operates with minimum head loss,accuracy:0.25%	12
5	Orifice Plate: 30mm diameter	13
6	Rotameter: Accuracy +/-2% of Full scale	14
7	pH meter: Portable pH meter range om 0 to as solution 0.1/0.01 pH	15
8	Data Acquisition System for any physical parameter monitoring and PC standard interface or any equivalent	16
9	Dual trace CRO with probe: Bandwidth AC 10Hz~20MHz (-3dB). DC 20MHz (-3dB), X10 Probe	2,3,4
10	Function generator: Frequency Ranges: 0.1 Hz to 1 MHz, output waveforms: sine, triangle, square, ramp, pulse	2,3,4,5,6
11	Digital storage oscilloscope: Bandwidth 60MHz,dual channel, sampling rate 1 GS/sec	5
12	Spectrum analyzer: 9 kHz - 6.2 GHz, Max acquisition BW:40MHz	6
13	Digital multi-meter: 0-10A, 0-600V, 0-10 Mohms with 3 -1/2 digital LCD display	7
14	DC power supply (0-30V), 1Amp.	7
15	LVDT: Stroke range ±0.1 plus/minus 2.54 or available range	8
16	Strain gauge: Universal general - purpose strain gauge	9

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Fundamentals of electronic measurements	CO1	4	0	0	0	0
2	II	Testing and measuring Instrument	CO2	8	0	0	0	0
3	III	Sensors and Transducers	CO3	6	0	0	0	0
4	IV	Application of Sensors and Tranducers	CO4	8	0	0	0	0
5	V	Data Acquisition System	CO5	4	0	0	0	0
Grand Total				**30**	**0**	**0**	**0**	**0**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Formative assessment of laboratory learning 50 marks.
- Each practical will be assessed considering 60% weigtage to process, 40% weightage to product.

Summative Assessment (Assessment of Learning)

- End semester summative Assessment each of 25 marks for laboratory learning.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	_	_	2	1	1	1			
CO2	2	_	1	2	1	2	2			
CO3	2	_	1	2	1	1	1			
CO4	2	_	1	2	2	2	2			
CO5	2	2	2	2	2	2	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Sawhney, A.K.	Electrical & Electronic Measurements & Instrumentations	Dhanpat Rai & sons, New Delhi, 2017 or latest edition 2021.
2	Kalsi,H.S.	Electronic Instrumentation	McGraw Hill, New Delhi,2010 ISBN:13-9780070702066 or latest edition

Sr.No	Author	Title	Publisher with ISBN Number
3	David,A.Bell	Electronic Instrumentation and Measuremens	Oxford University Press, New Delhi edition- 2013 ISBN : 10:0-19-569614-X or latest edition
4	Helfrick, A.D. Cooper, W.D.	Modern Electronic Instrumentation and measurement Techniques	Pearson Education India,1st Edition, New Delhi,2015,ISBN-13:978 or latest edition
5	Ghosh, A.K	Introduction to Measurement & Instrumentation	Prentice Hall India Learning Private Limited 2013, 4th or latest Edition, ISBN-13-978-8120346253.
6	Murty, D.V.S.	Transducers and Instrumentaion	Prentice Hall India Learning Private Limited, 2nd Edition, ISBN-13-978-8120335691
7	Patranabis D.	Sensors and Transducers	PHI Learning Private Limited, 2nd Edition, ISBN-978-81-203-2198-4

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://en.wikipedia.org/wiki/Electronic_test_equipment	Testing using CRO, multimeter, function generator and spectrum analyser.
2	https://en.wikipedia.org/wiki/List_of_electrical_and_electronic_measuring_equipment	Measuring instruments used in electrical and electronic work.
3	https://en.wikipedia.org/wiki/Instrumentation	Sensors to measure physical and electrical quantities
4	https://en.wikipedia.org/wiki/Flow_measurement	Flowmeters
5	https://en.wikipedia.org/wiki/Data_acquisition	Data acquisition system

Sr.No	Link / Portal	Description

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students